Spatial **prediction** of **wildfire** susceptibility using field **survey** GPS data and machine learning approaches

O Ghorbanzadeh, K Valizadeh Kamran, T Blaschke… - Fire, 2019 - mdpi.com

… A **wildfire** data inventory was created by integrating global …) approaches for the spatial **prediction** of **wildfire** susceptibility. The … All ML approaches were trained using 75% of the **wildfire** …

☆ Save 99 Cite Cited by 173 Related articles All 13 versions »

Physical **model** of **wildland fire spread**: Parametric uncertainty analysis

X Yuan, N Liu, X Xie, DX Viegas - Combustion and Flame, 2020 - Elsevier

… the **prediction** of **spread** rate may arise from uncertain, imprecise or improper determinations of the **model** … is made on a typical **model** of upslope fire **spread** that is consistent with many …

☆ Save 99 Cite Cited by 33 Related articles All 2 versions

Progress and prospects for predicting **wildfire spread** through the wildland-urban interface

O Price, S Ondei, DMJS Bowman - International Journal of Disaster Risk …, 2025 - Elsevier

… Scientific understanding of **wildfire spread** into urban areas is … **review** the state of knowledge around modelling fire **spread** in … We suggest that our proposed **model** is feasible given …

☆ Save 99 Cite Related articles All 2 versions

Prediction of **wildfire** rate of **spread** in grasslands using machine learning methods

S Khanmohammadi, M Arashpour… - … Modelling & Software, 2022 - Elsevier

… **Prediction** of **wildfire propagation** plays a crucial role in reducing … models to predict the rate of **spread** of grassfires. A dataset from … These models produced excellent fits to the **model** …

☆ Save 99 Cite Cited by 44 Related articles All 6 versions

Statistical models of key components of **wildfire** risk

DDZ Xi, SW Taylor, DG Woolford… - … **Review** of Statistics and …, 2019 - annualreviews.org

… It is important to note that not all **wildland fire** ignitions may … fire **spread** may be detected by **wildland fire** management … Improving **prediction** of daily **wildfire dynamics** is critical for …

☆ Save 99 Cite Cited by 72 Related articles All 6 versions

On the need for a theory of **wildland fire spread**

MA Finney, JD Cohen, SS McAllister… - … journal of **wildland fire**, 2012 - CSIRO Publishing

… In summary, our **review** of existing research indicates an insufficient experimental basis for … an insufficient basis for related fire **spread model** assumptions. Independent experimental …

☆ Save 99 Cite Cited by 310 Related articles All 12 versions

Representation and evaluation of **wildfire propagation** simulations

JB Filippi, V Mallet, B Nader - … Journal of **Wildland Fire**, 2013 - CSIRO Publishing

… a **wildfire simulation**, reviews the most common scoring methods using this formalism, and proposes new methods that are explicitly designed to evaluate a **forest fire simulation** … **review** …

☆ Save 99 Cite Cited by 72 Related articles All 11 versions

Wildfire risk modeling

S Oliveira, J Rocha, A Sá - Current Opinion in Environmental Science & …, 2021 - Elsevier

… **review** of the methods and approaches applied to **model wildfire** risk and its key components, in particular since 2018, and how they can contribute to **wildfire** management. …

☆ Save 99 Cite Cited by 76 Related articles All 5 versions

Predicting **wildfire spread** and behaviour in Mediterranean landscapes

M Salis, B Arca, F Alcasena… - … of **Wildland Fire**, 2016 - CSIRO Publishing

… Nevertheless, to incorporate fire **spread simulation** in operational workflows, it is crucial to increase the **model** accuracy by suitable validation. The validation of modelled fire behaviour …

☆ Save 99 Cite Cited by 77 Related articles All 8 versions

Forest fire spread simulation and fire extinguishing visualization research

Q Meng, H Lu, Y Huai, H Xu, S Yang - Forests, 2023 - mdpi.com

… **forest fire spread** quickly. Therefore, this paper proposes a lightweight **forest fire spread model** … ensuring the realism of **forest fire behavior** and the immersion of the virtual environment. …

☆ Save 99 Cite Cited by 17 Related articles All 5 versions »